2024

COMPUTER SCIENCE

Paper : CSMC-101

(Mathematics for Computing)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1** and **2**, and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) Define block of a graph.

(b) Compare between the maximum independent set and a maximal independent set.

(c) If the length of a Prüfer sequence P is L, then how many vertices will be there in the corresponding tree?

(d) What would be the Chromatic number of a Cycle G with an odd number of nodes? Why?

(e) Is the set $w = \{(x, y, z) \in R^3 : x - 2y + 3z = 0\}$ a subspace of the real vector space R^3?

(f) Define "inner product" of two vectors of a real vector space.

(g) Define the basis of a vector space.

(h) State why the following matrices are not in REF. Use elementary row operations to put them in REF.

$$M = \begin{bmatrix} 3 & -1 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \end{bmatrix}$$

2. Answer ***any five*** questions : 4×5

(a) "K_5 is a planar graph."— Comment on the correctness of the statement and justify your opinion.

(b) "Two blocks in a graph share at most one vertex."— Comment on the correctness of the statement and justify your opinion.

(c) "A 6-vertex graph cannot be self-complementary."— Comment on the correctness of the statement and justify your opinion.

(d) What form does a particular solution of the linear non-homogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$ have for $F(n) = n^2 2^n$ and for $F(n) = n3^n$?

Please Turn Over

(e) Find the algebraic and geometric multiplicities of each eigenvalue of the following matrix.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{bmatrix}$$

(f) Find the linearly independent subset T of the set S = { (1, 2, −1), (−3, −6, 3), (2, 1, 3), (8, 7, 7)} belongs to R^3 which spans the same space as S.

(g) Prove that an orthogonal set of non-null vectors in a Euclidian space V is linearly dependent.

(h) Investigate for what values of a,b the following system of equations x + y + z = 6, x + 2y + 3z = 10, x + 2y + az = b has (i) only one solution (ii) no solution (iii) infinitely many solutions.

3. (a) Find the generating functions for the following :

(i) $h_n = 2h_{n-1} + 3^n,\ h_0 = 0$

(ii) $h_n = 3h_{n-1} + 4h_{n-2},\ h_0 = h_1 = 1.$

(b) For each of the above, use the generating functions to find a non-recursive formula for h_n.

(c) Find the solution of the recurrence relation $a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3$ with $a_0 = 1$ and $a_1 = 4$. 3+3+4

4. (a) Write an algorithm to encode any tree with N nodes into a Prüfer sequence P.

(b) Illustrate and justify with an example with at least 6 vertices in a spanning tree that for *every tree there is exactly one corresponding Prüfer Sequence, and for each Prüfer Sequence there is exactly one corresponding tree.* 4+6

5. A family has three children— each of them is either a boy or a girl.

(a) What will be the cardinality of the sample space for this with all possibilities? Write the sample space.

(b) Determine whether the following pair of events are mutually exclusive.

M : {The family has at least one boy}

N : {The family has all girls}.

(c) Find the conditional probability of having two boys and a girl, given that the firstborn is a boy. (1+2)+2+5

6. (a) Given the following matrix. Find a basis of the row space of A. Also, show that the row rank of A = Column rank of A.

(b) Determine whether the three vectors lie on the same line in R^3.

v1 = (2, −1, 4), v2 = (4, 2, 3), v3 = (2, 7, −6)

(c) What do you mean by the dimension of a vector space? 6+3+1

7. (a) Prove that for any two vectors u, v in a Euclidian space w (real inner product space).

$|< u, v >| \leq || u || . || v ||$ and the equality holds when u and v are linearly dependent.

(b) Determine if A is diagonalizable. If yes, find a matrix P that diagonalizes A.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 2 \\ -2 & 0 & 1 \end{bmatrix}$$

5+5

8. (a) Suppose $\{u_1, u_2, u_3, u_4\}$ is a basis of vector space V over a field F and a non-zero vector w of V is expressed as $w = a_1u_1 + a_2u_2 + a_3u_3 + a_4u_4$; where $a_i \in F(i = 1, 2, 3, 4)$ then if a_2 not equals to 0; prove that $\{u_1, u_2, u_3, u_4\}$ is a new basis.

(b) Prove that if vl, v2, ...vp are the vectors in V, then span {vl, v2,.....vp} is a subspace of V.

6+4